# 11 PHYSICS

# MOTION & FORCES

# VECTORS

**Scalar quantities** - have magnitude or size only.

e.g. mass, distance, speed, density, temperature.

**Vector quantities** - have magnitude and direction.

e.g. displacement, velocity, acceleration, force, impulse, momentum.

Vectors are represented by arrows - the length indicates the magnitude of the vector and the arrowhead indicates the direction in which it acts.

i.e.

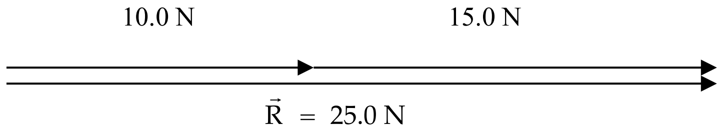

## COMBINING VECTORS

**Addition**

| |
|---|
| Vectors are added head-to-tail. The resultant vector runs from the tail of the first vector to the head of the last vector. |

1. <u>**Addition**</u>

**One-dimension**

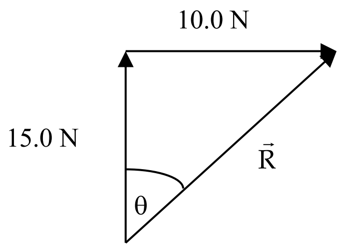



The resultant vector R = 25 .0 N - it runs from the tail of the first vector to the head of the last vector.

**Two Dimensions**

**Vectors At Right Angles**

Use Pythagorus' Theorem and trigonometry to find the resultant.



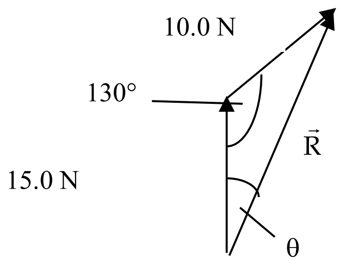

$$\vec{R} = \sqrt{(10.0)^2 + (15.0)^2}$$
$$= 18.03 \text{ N}$$

$$\tan\theta = \frac{10.0}{15.0}$$
$$\Rightarrow \quad \theta = 33.69°$$

$\therefore$ <u>$\vec{R}$ = 18.0 N at 33.7° to the 15.0 N vector.</u>

**Vectors Not At Right Angles**

Use the sine and cosine rules for triangles to find the resultant vector.




$$\vec{R} = \sqrt{(15.0)^2 + (10.0)^2 - 2(15.0)(10.0)\cos 130°}$$

$$= 22.76 \text{ N}$$

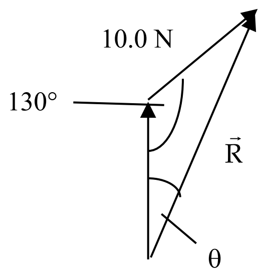
$$\frac{10.0}{\sin\theta} = \frac{22.76}{\sin 130°}$$

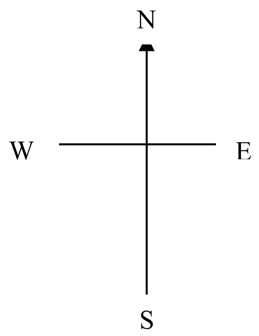
$$\Rightarrow \theta = 19.67°$$

$\therefore$ $\underline{\vec{R} = 22.8 \text{ N at } 19.7° \text{ to the } 15.0 \text{ N vector}}$.

**Note:** If compass directions are given, it makes it easier to describe the direction.

Consider the example above.




The resultant vector R is at a 19.7° angle east of north.

$\therefore$ Direction is N 19.7° E.

Also, north can be considered the 0° bearing, with all other bearings determined clockwise from this direction.

$\therefore$ Direction is a bearing of 19.7°.

<u>Hence R = 22.8 N N 19.7° E or R = 22.8 N on a bearing of 19.7°</u>.

**Parallelogram Method**

If two vectors are acting at the same point, it is easiest to use the ***parallelogram method*** to add them.

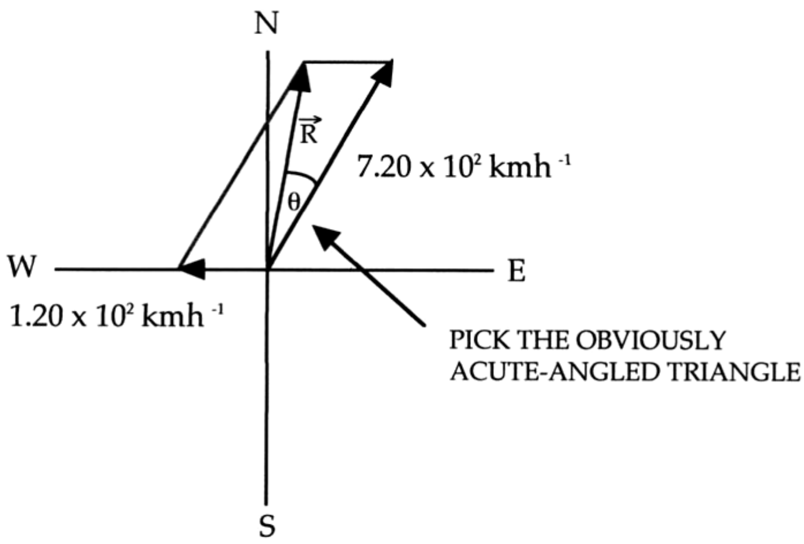
**Note:** Even if the vectors don't act at the same point, this method proves to be the best for solving problems.

The "rules" for using this method are simple.

1. Draw in a set of "cross-hairs" (with compass directions if appropriate).
2. Draw the two vectors originating at the centre. Make their size ***roughly to scale***.
3. Construct the other two sides of the parallelogram.
4. Rule in the resultant vector, which is the ***diagonal starting at the centre***.
5. Of the two triangles formed, choose the one with an ***obviously acute angle*** near the centre of the cross-hairs.

**Example**

A plane flying at $7.20 \times 10^2$ kmh$^{-1}$ N30.0°E has to contend with a wind of $1.20 \times 10^2$ kmh$^{-1}$ from the east. Determine the resultant velocity of the plane over the ground.




$$\vec{R} = \sqrt{(7.20 \times 10^2)^2 + (1.20 \times 10^2)^2 - 2(7.20 \times 10^2)(1.20 \times 10^2)\cos 60.0°}$$
$$= 6.681 \times 10^2 \text{ kmh}^{-1}$$
$$= 1.856 \times 10^2 \text{ ms}^{-1}$$

To find θ: $\dfrac{6.681 \times 10^2}{\sin 60.0°} = \dfrac{1.20 \times 10^2}{\sin \theta}$

$\Rightarrow \quad \theta = 8.949°$

$\therefore \quad \underline{\vec{R} = 1.86 \times 10^2 \text{ ms}^{-1} \text{ N21.0°E}}$

2. <u>**Subtraction**</u>

**One-dimension**

The easiest way to deal with vectors in one dimension is to ***choose one direction as positive***.

Any vector in this direction is positive while all those in the other direction are negative.

It is then a simple matter of adding numbers.

**Example 1**
A car moving at 15.0 $ms^{-1}$ brakes suddenly to avoid a dog and reduces to a velocity of 5.00 $ms^{-1}$. Calculate its change in velocity.

Take the forwards direction as positive.

By definition the change in velocity is given by :

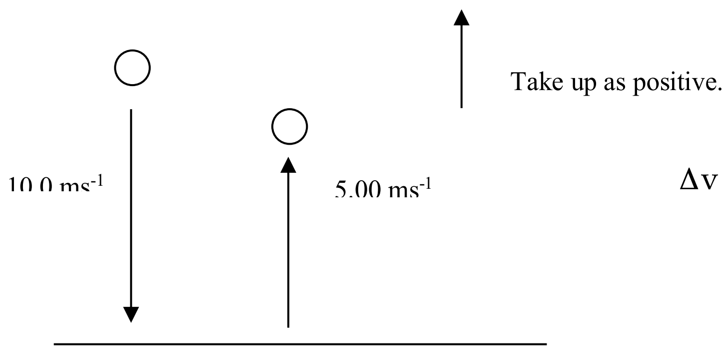
$$\Delta v = v - u$$
$$= 5.00\ ms^{-1} - 10.0\ ms^{-1}$$
$$= -5.00\ ms^{-1}$$

$\therefore$ Change in velocity = 5.00 $ms^{-1}$ backwards.

The answer is negative which means the resultant is in the opposite direction to the one you chose as positive.

i.e. The direction is backwards.

**Example 2**
A ball falls onto a concrete floor at 10.0 $ms^{-1}$ and rebounds at 5.00 $ms^{-1}$. Calculate the change in velocity of the ball.




$u = -10.0\ ms^{-1}$ since it is down in the negative direction.

$$\Delta v = v - u$$
$$= 5.00 - (-10.0)$$
$$= \underline{15.0\ ms^{-1}\ \text{upwards}}$$

Answer is positive so the final direction is upwards in the positive direction chosen.

2. **Two-dimensions**

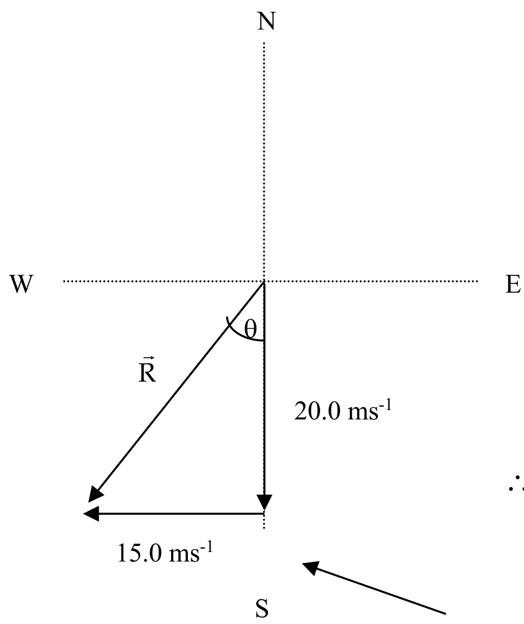
When subtracting vectors, the "negative" sign means:

***"change the direction of the subtracted vector 180° and add it to the other vector"***.

This works best when vectors act at an angle to each other.

**Example**

A car moving at 20.0 $ms^{-1}$ north makes a left-hand turn around a sweeping bend and heads west at 15.0 $ms^{-1}$. Calculate its change in velocity.




$$\Delta v = v - u$$

$$= 15.0\ ms^{-1}\ W - 20.0\ ms^{-1}\ N$$

$$= 15.0\ ms^{-1}\ W + 20.0\ ms^{-1}\ S$$

$$\vec{R} = \sqrt{(15.0)^2 + (20.0)^2} \qquad \tan\theta = \frac{15.0}{20.0}$$

$$= 25.0\ ms^{-1} \qquad \Rightarrow \quad \theta = 36.87°$$

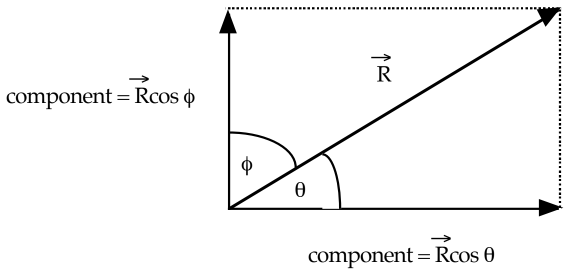
$$\therefore \quad \underline{\Delta v = 25.0\ ms^{-1}\ S\ 36.9°W}$$

The use of a compass grid line makes it easier to determine the direction of the resultant vector.

## COMPONENTS OF A VECTOR

Any single vector can be expressed as ***two component vectors that are at right angles to each other***.

***<u>Neither component vector can be bigger than the original vector</u>***.




Any component is given by:

$$\boxed{\textbf{component} = \vec{\mathbf{R}}\mathbf{cos}\ \theta}$$

where R = given vector.
θ = the angle between R and the direction of the required component.

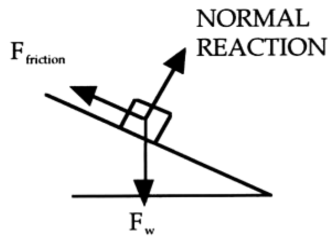
**Examples**

1. **Inclined Plane**

A box weighing 80.0 N sits at rest on a plane inclined at 27.0° to the horizontal.

(a) Draw a diagram showing all forces acting on the box.
(b) Calculate the force exerted perpendicular to the plane by the box onto the plane.

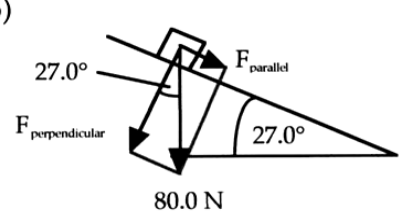
(a)




Only three forces act on the box.

The components of $F_w$ are **not** forces that are acting.

(b)




$F_{perp.} = F_w \cos 27.0°$
$= 80.0 \cos 27.0°$
$= 71.28$ N

∴ <u>$F_{perp.}$ = 71.3 N into the plane.</u>

2.

**A box of mass 12.5 kg is sliding down a frictionless plane, inclined at 35° to the horizontal. Determine the force.**

**(a) (i) acting down the plane**

**(ii) into the plane**

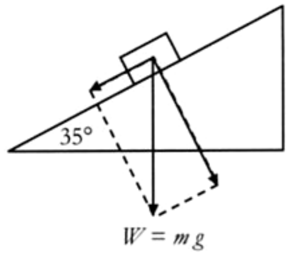
**(b) Assuming $g = 9.80\ \text{ms}^{-2}$, determine the acceleration of the mass down the plane.**

(a) Force $\|$ to the plane

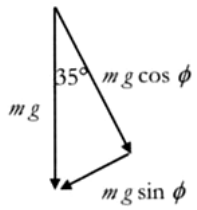
$$F_{\|} = m\,g\sin\theta = (12.5)(9.80)(\sin 35°) = 70.3\ \text{N}$$

Force perpendicular to the plane

$$F_{perp} = m\,g\cos\theta = (12.5)(9.80)(\cos 35°) = 100\ \text{N}$$




or




(b) Since $F = m\,a$

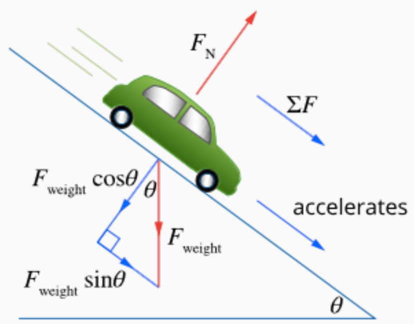
$$a = \frac{F}{m}$$

$\|$ to plane

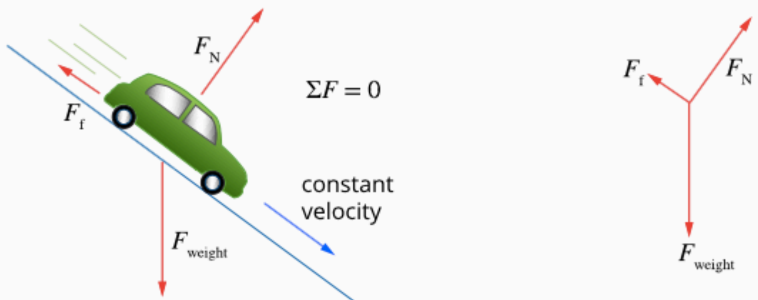
$$a = \frac{mg\sin\theta}{m}$$

$$a = (9.8)\ (\sin 35) = 5.62\ \text{ms}^{-2}$$

3. Consider the forces acting on a smooth incline (no friction) and one with friction.




**2.1.4** • The motion of an object on a smooth inclined plane can be analysed by finding the components of the weight force that are parallel and perpendicular to the plane.

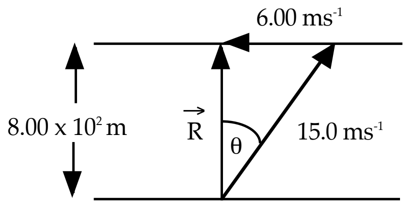



**2.1.5** • If friction acts on the car, causing it to move with a constant velocity, the net force on the car will be zero, and so the forces will be in balance both perpendicular and parallel to the incline.

4. **River Problem**

A person wants to cross a river $8.00 \times 10^2$ m wide to reach a jetty directly across on the opposite bank. If the river is flowing at $6.00\ ms^{-1}$ and the boat that is used can travel at $15.0\ ms^{-1}$ in still water, calculate:

(a) the resultant velocity of the boat in relation to the river bottom (or bank).
(b) the time taken to reach the other side.




(a) $\vec{R} = \sqrt{(15.0)^2 - (6.00)^2}$
$= 13.75\ ms^{-1}$

$\therefore$ $\underline{\vec{R} = 13.8\ ms^{-1} \text{ at } 90.0° \text{ to the bank.}}$

(b) To calculate the time taken, use a component of any velocity that is directly across the river.

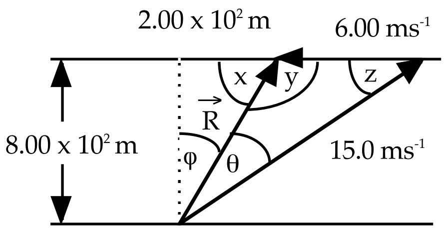
In this example, $\vec{R}$ is directly across so use that.

i.e. $v_{across} = \dfrac{s_{across}}{t_{across}}$

$\Rightarrow$ $t = s / v$
$= 8.00 \times 10^2 / 13.75$
$= 58.18$ s

$\therefore$ Time taken = 58.2 s .

5. If the person in Question 3 was trying to reach a jetty $2.00 \times 10^2$ m upstream, calculate the resultant velocity and the time taken.




$\tan \phi = 2.00 \times 10^2 / 8.00 \times 10^2$

$\Rightarrow \phi = 14.04°$

By geometry: $x = 75.96°$

$y = 104.04°$

To find θ: $\dfrac{6.00}{\sin \theta} = \dfrac{15.0}{\sin 104.04°}$

$\Rightarrow \theta = 22.83°$

By geometry: $z = 53.13°$

To find R: $\dfrac{\vec{R}}{\sin 53.13°} = \dfrac{6.00}{\sin 22.83°}$

$\Rightarrow \vec{R} = 12.37 \text{ ms}^{-1}$

∴ $\vec{R}$ = 12.4 $\text{ms}^{-1}$ at 53.1° to the bank upstream.

To calculate the time taken to cross, use any component of velocity that is directly across the river.

e.g. 12.37 cos 14.04° or 15.0 cos 36.87°

$$v_{across} = \frac{s_{across}}{t_{across}}$$

$$\Rightarrow \quad t = s / v$$
$$= 8.00 \times 10^2 / 12.37 \cos 14.04°$$
$$= 66.66 \text{ s}$$

∴ Time taken = 66.7 s .

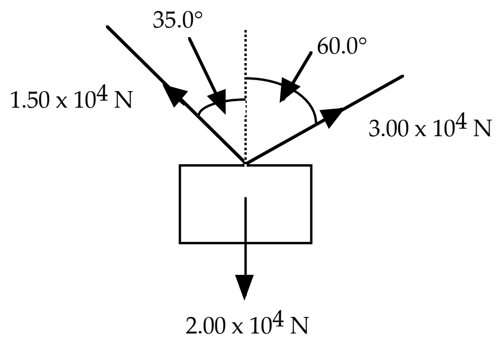
# ADDING MORE THAN TWO VECTORS

The best way to solve these problems is to use ***components*** to reduce the problem to the addition of two vectors at right angles to each other.

Generally this will be done by considering movement in two dimensions within a plane.
i.e. up / down /left / right
north /south / east / west (if compass directions are given).

**Example**

A heavy crate is being lifted by two cables as shown below.



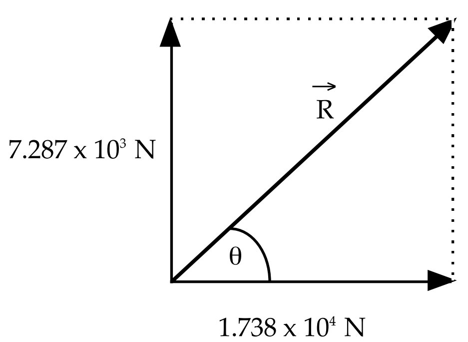

If the crate weighs 2.00 x $10^4$ N and the tension in each cable is as shown, calculate the resultant force experienced by the crate.

VERTICALLY Take up as +ve.

$$F_v = 1.50 \times 10^4 \cos 35.0° + 3.00 \times 10^4 \cos 60.0° - 2.00 \times 10^4$$
$$= 7.287 \times 10^3 \text{ N}$$

HORIZONTALLY Take to the right as +ve.

$$F_h = 3.00 \times 10^4 \cos 30.0° - 1.50 \times 10^4 \cos 55.0°$$
$$= 1.738 \times 10^4 \text{ N}$$




$$\vec{R} = \sqrt{(7.287 \times 10^3)^2 + (1.738 \times 10^4)^2}$$
$$= 1.885 \times 10^4 \text{ N}$$

$$\tan \theta = 7.287 \times 10^3 \ / \ 1.738 \times 10^4$$
$$\Rightarrow \theta = 22.75°$$

$\therefore$ $\vec{R}$ = 1.88 x$10^4$ N at 22.8° above the horizontal and to the right.

## MOTION IN ONE DIMENSION

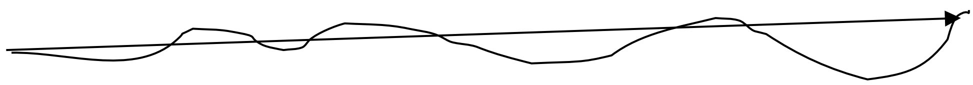
**Definitions**

**Displacement (s)**: change of position of a body in a given direction.
i.e. a vector

**Distance**: how far a body moves in any direction.
i.e. a scalar

**Example**

A car moves along West Coast Drive from Sorrento to Trigg. Its motion is represented in the diagram below.



The distance on the road is 5.30 km. (The direction is not important and the distance is the sum total of all movements.)

The displacement (measured from the start of the motion to the finish) is 4.75 km S 5.0° E.

**Velocity (v)**: the rate of change of position in a given direction.
i.e. a vector

$$\boxed{\mathbf{v} = \frac{\mathbf{s}}{\mathbf{t}}}$$

where v = the average velocity for the motion ($ms^{-1}$)
s = the displacement of the object or body (m)
t = the time taken for the entire motion (s)

**Speed**: the rate of change of distance.
i.e. a scalar

$$\boxed{\textbf{speed} = \frac{\mathbf{d}}{\mathbf{t}}}$$

where d = the total distance travelled by the object (m)
t = time taken for the entire movement (s)

**Example**

A competitor in a rogaining competition covers 2.50 x $10^2$ m south in 65.0 s, 3.50 x$10^2$ m west in 1.20 x $10^2$ s and 4.00 x $10^2$ m north in 1.05 x $10^2$ s. Calculate:

(a) the distance covered in this time.
(b) the displacement of the competitor.
(c) the average speed maintained.
(d) the average velocity of the competitor.

(a) $d = 2{\cdot}50\times10^2 + 3{\cdot}50\times10^2 + 4{\cdot}00\times10^2$

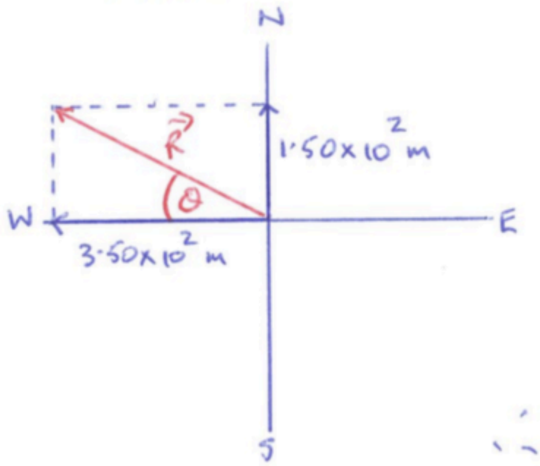
$= \underline{1{\cdot}00\times10^3 \text{ m}}$

(b) Consider the north-south direction - take N as +ve

$\Rightarrow s = 4{\cdot}00\times10^2 - 2{\cdot}50\times10^2$

$= 1{\cdot}50\times10^2 \text{ m N.}$

Add this vector to $3{\cdot}50\times10^2$ m W




$\vec{R} = \sqrt{(1{\cdot}50\times10^2)^2 + (3{\cdot}50\times10^2)^2}$

$= 3{\cdot}81\times10^2 \text{ m}$

$\tan\theta = \dfrac{1{\cdot}50\times10^2}{3{\cdot}50\times10^2}$

$\Rightarrow \theta = 23{\cdot}2^\circ$

$\therefore \underline{s = 3{\cdot}81\times10^2 \text{ m W } 23{\cdot}2^\circ \text{ N}}$

(c) speed $= \dfrac{d}{t}$

$= \dfrac{1{\cdot}00\times10^3}{2{\cdot}90\times10^2}$ ← Calculate the total time

$= \underline{3{\cdot}45 \text{ ms}^{-1}}$

(d) $v_{av} = \dfrac{s}{t}$

$= \dfrac{3{\cdot}81\times10^2}{2{\cdot}90\times10^2}$

$= \underline{1{\cdot}31 \text{ ms}^{-1} \text{ W } 23{\cdot}2^\circ \text{ N}}$ ← Must include the direction

**Acceleration (a)**: the rate of change of velocity in a given direction.
i.e. a vector

$$\mathbf{a} = \frac{\Delta \mathbf{v}}{\mathbf{t}} = \frac{\mathbf{v - u}}{\mathbf{t}}$$

where
a = the acceleration of the object ($ms^{-2}$)
v = final velocity of the object ($ms^{-1}$)
u = initial velocity of the object ($ms^{-1}$)
t = time taken for the acceleration to occur (s)

For uniformly accelerated motion, the following equation can be used where appropriate.

$$v_{ave} = \frac{u + v}{2}$$

where $v_{ave}$ = the average velocity for the accelerated motion ($ms^{-1}$)

**Example**
A car moving at 60.0 $kmh^{-1}$ along a level road brakes sharply to avoid a ball that has rolled down a driveway. It decreases uniformly to 20.0 $kmh^{-1}$ in 2.60 s. Calculate the acceleration experienced by the car.

Take forwards as +ve

$v = 20.0\ kmh^{-1} = 5.56\ ms^{-1}$

$u = 60.0\ kmh^{-1} = 16.7\ ms^{-1}$

$a = ?$

$t = 2.60\ s$

$s = ?$

$$a = \frac{v-u}{t}$$

$$= \frac{5.56 - 16.7}{2.60}$$

$$= -4.28\ ms^{-2}$$

$\therefore$ $\underline{a = 4.28\ ms^{-2} \text{ backwards}}$

**EQUATIONS OF MOTION**

For objects and bodies undergoing uniform acceleration, a series of equations can be derived to describe the motion.

$$\boxed{\begin{aligned} v &= u + at \\ s &= ut + \tfrac{1}{2}at^2 \\ v^2 &= u^2 + 2as \end{aligned}}$$

where
- v = final velocity ($ms^{-1}$)
- u = initial velocity ($ms^{-1}$)
- a = aceleration ($ms^{-2}$)
- t = time taken for the acceleration (s)
- s = displacement during the acceleration ($ms^{-2}$)

These equations can be used to solve problems involving objects moving horizontally across the ground, down inclined planes such as hills, or vertically under the influence of Earth's gravity.

Near the Earth's surface, the acceleration due to gravity is accepted to be:

$$\mathbf{g = 9.80\ ms^{-2}}.$$

**Example 1**
A car moving at 60.0 $kmh^{-1}$ along a level road brakes sharply to a stop in 2.70 s, in order to miss a ball that had rolled down a driveway. Calculate the distance covered by the car, assuming it was in a straight line.

Take forwards as +ve.

$v = 0\ ms^{-1}$
$u = 16.7\ ms^{-1}$
$a = ?$
$t = 2.70s$
$s = ?$

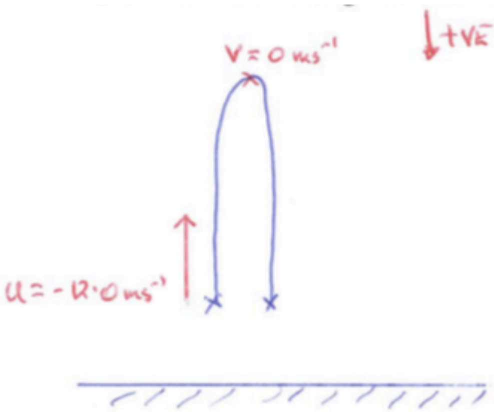
$$a = \frac{v-u}{t} = \frac{0-16.7}{2.70} = -6.18\ ms^{-2}$$

$$s = ut + \tfrac{1}{2}at^2 = (16.7)(2.70) + \tfrac{1}{2}(-6.18)(2.70)^2 = \underline{22.6\ m\ \text{forwards}}$$

$$v^2 = u^2 + 2as \Rightarrow s = \frac{v^2 - u^2}{2a} = \frac{0-(16.7)^2}{2(-6.18)} = \underline{22.6m\ \text{forwards}}$$

**Example 2**
A boy playing catch by himself throws a ball vertically upwards at 12.0 ms$^{-1}$ and catches it at the same height as the release point. Calculate:

(a) the maximum height reached above the release point.
(b) the time of flight for the ball.




(a) Consider movement to the top.

$v = 0\ ms^{-1}$
$u = -12.0\ ms^{-1}$
$a = 9.80\ ms^{-2}$
$t = ?$
$s = ?$

$$v^2 = u^2 + 2as$$
$$\Rightarrow s = \frac{v^2 - u^2}{2a}$$
$$= \frac{0 - (-12.0)^2}{2(9.80)}$$
$$= -7.35\ m$$

$\therefore$ Height above release = 7.35 m

(b) Consider movement to the top.

$$v = u + at$$
$$\Rightarrow t = \frac{v-u}{a}$$
$$= \frac{0-(-12.0)}{9.80}$$
$$= 1.22\ s$$

$$\text{Total time} = 2 \times 1.22$$
$$= 2.44\ s$$

OR Consider the whole movement.

$v = ?$
$u = -12.0\ ms^{-1}$
$a = 9.80\ ms^{-2}$
$t = ?$
$s = 0\ m$

$$s = ut + \tfrac{1}{2}at^2$$
$$\Rightarrow 0 = -12.0t + \tfrac{1}{2}(9.80)t^2$$
$$\Rightarrow 4.90\cancel{t^2} = 12.0\cancel{t}$$
$$\Rightarrow t = 2.45\ s$$

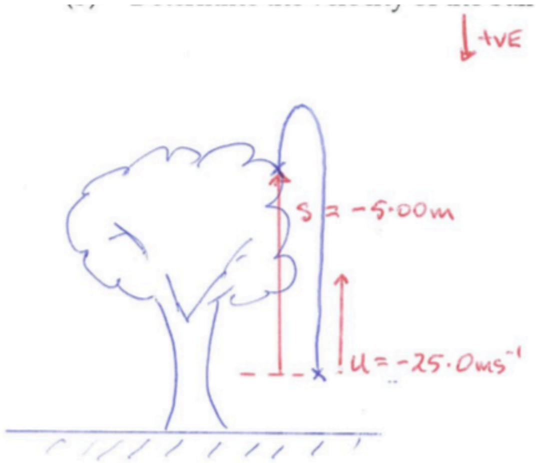
NOTE: The difference between the two times is due to rounding.

**Example 3**

A girl playing with a racquet and ball hits the ball vertically upwards at 25.0 ms$^{-1}$. It rises above a tree and gets caught by a branch on the way down. It finishes 5.00 m above the point at which it was struck.

(a) Calculate the time of flight for the ball.

(b) Determine the velocity of the ball as it hits the branch.




(a) Consider the whole motion.

$v = ?$

$u = -25.0\,\text{ms}^{-1}$

$a = 9.80\,\text{ms}^{-2}$

$t = ?$

$s = -5.00\,\text{m}$

$s = ut + \frac{1}{2}at^2$

$\Rightarrow -5.00 = -25.0t + \frac{1}{2}(9.80)t^2$

$\Rightarrow 4.90t^2 - 25.0t + 5.00 = 0$

$$\Rightarrow t = \frac{25.0 \pm \sqrt{(-25.0)^2 - 4(4.90)(5.00)}}{2(4.90)}$$

$\Rightarrow t = 4.89$ or $\cancel{0.208}$ s  Unrealistic answer

$\therefore$ $\underline{t = 4.89\,\text{s}}$

OR Find the velocity first.

i.e. $v^2 = u^2 + 2as$

$= (-25.0) + 2(9.80)(-5.00)$

$\Rightarrow v = 23.0\,\text{ms}^{-1}$ down.

$v = u + at$

$$\Rightarrow t = \frac{v-u}{a}$$

$$= \frac{23.0 - (-25.0)}{9.80}$$

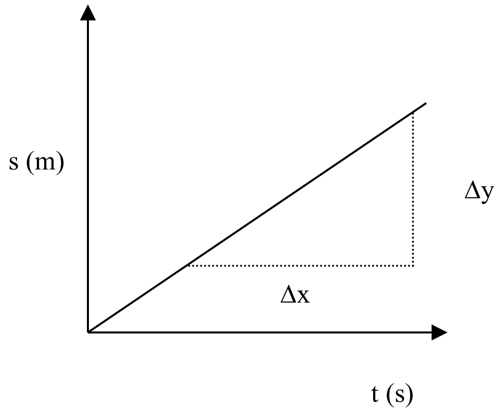
$= \underline{4.90\,\text{s}}$.  (Answer different due to rounding.)

(b) $\underline{v = 23.0\,\text{ms}^{-1}\text{ down}}$  (See working above)

# MOTION GRAPHS

## Displacement - Time

### Constant Velocity

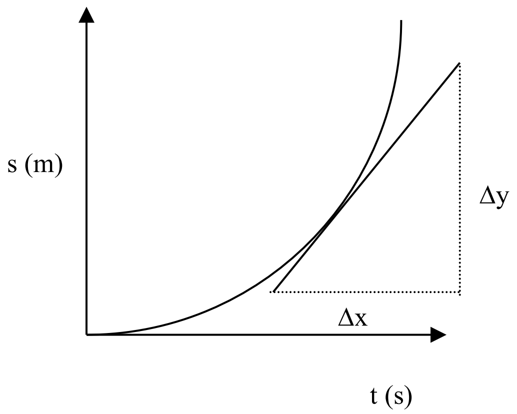



The slope or gradient of the line is given by:

$$\text{slope} = \frac{\Delta x}{\Delta y} = \frac{s}{t}$$

Hence the slope of the graph gives the ***constant velocity*** of the body or object.

### Constant Acceleration



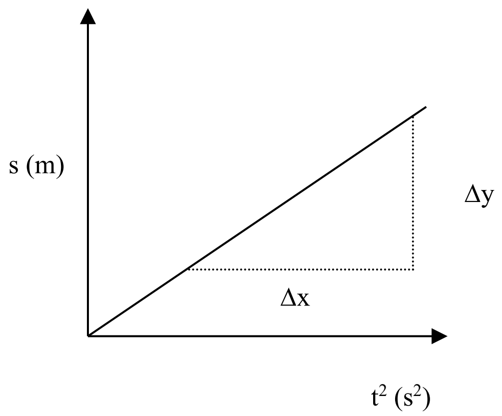

The graph at left represents an object falling under gravity or one that is uniformly accelerated across the ground.

The slope at a point on the curve gives the ***instantaneous velocity*** of the object at that time.

The graph is a parabola and clearly suggests that $s \propto t^2$.

The equation relating displacement and acceleration is $s = ut + 1/2\ at^2$.

If $s \propto t^2$, then plotting s versus $t^2$ should produce a straight line graph.




The slope or gradient represents 1/2 a.

The intercept represents ut.

The general equation of a line is:

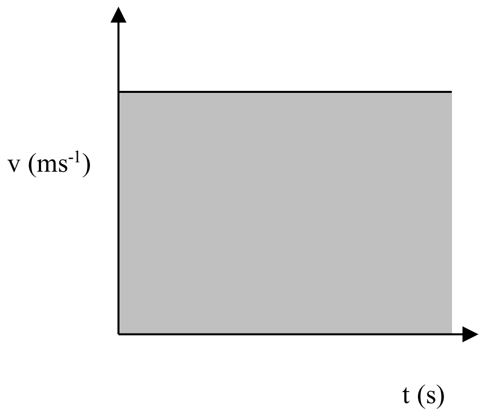
$$y = mx + b$$

b represents ut while m = 1/2 a.

**Important:** The slope of a linear graph generally means something - it is related to an equation that is applicable. ***You must know what the slope or gradient represents!***

**Velocity - Time**

**Constant Velocity**

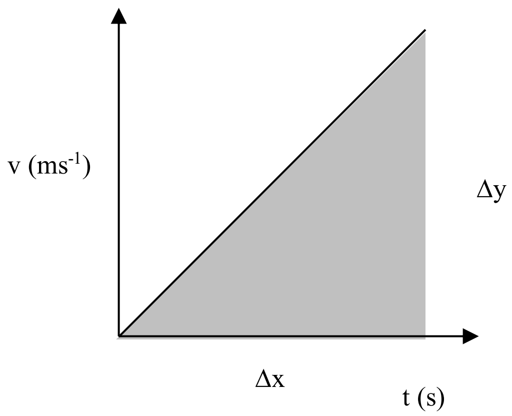



***The area under a v - t graph represents the displacement for the body.***

i.e. $s = v \times t$

**Constant Acceleration**

The slope gives the ***acceleration*** of the body.



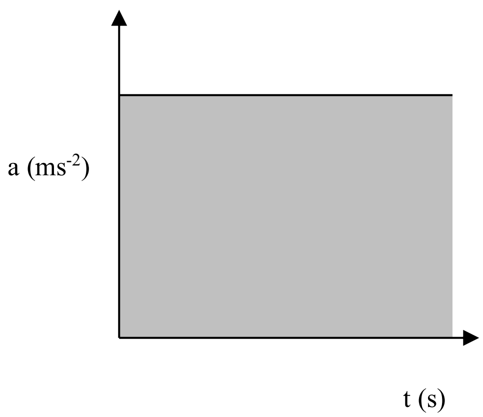

$$\text{slope} = \frac{\Delta v}{\Delta t} = a$$

Area under graph = displacement

= 1/2 x base x height

**Acceleration - Time**

**Constant Acceleration**



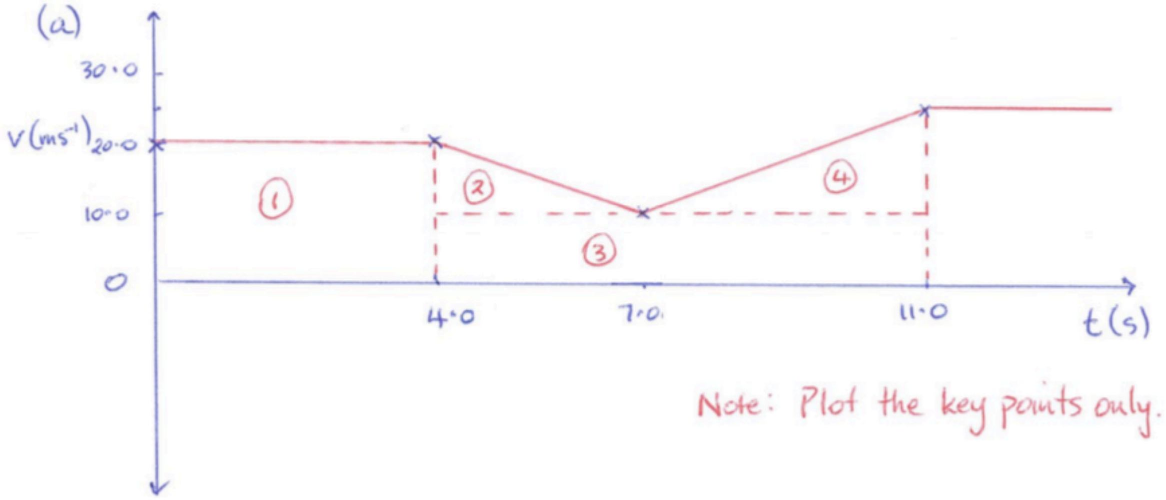

***Area under graph = change in velocity***

***= $\Delta v$***

**Example**

A car travels at 20.0 ms$^{-1}$ as it passes an observer on the side of the road. The car maintains this velocity for 4.00 s before decelerating uniformly for 3.00 s to 10.0 ms$^{-1}$. In the next 4.00 s it uniformly increases to 25.0 ms$^{-1}$ before continuing at constant velocity. The entire motion occurs in a straight line.

(a) Draw a velocity - time graph for the first 11.0 s of the motion.
(b) Calculate the displacement of the car in this time.
(c) Draw an acceleration - time graph for the motion.

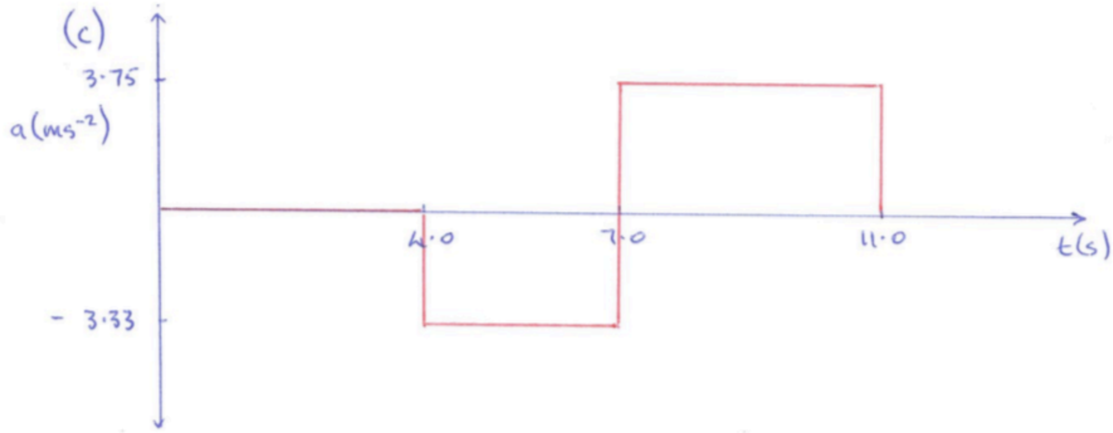



(b) $s$ = area under graph

$= (20.0 \times 4.00) + (\frac{1}{2} \times 3.00 \times 10.0) + (7.00 \times 10.0) + (\frac{1}{2} \times 4.00 \times 15.0)$

$= \underline{195 \text{ m}}$

(c)




Between $t = 4.0$ and $7.0 s$:

$$a = \frac{v-u}{t}$$
$$= \frac{10.0 - 20.0}{3.00}$$
$$= -3.33 \ ms^{-2}$$

Between $t = 7.0$ and $11.0 s$:

$$a = \frac{v-u}{t}$$
$$= \frac{25.0 - 10.0}{4.00}$$
$$= 3.75 \ ms^{-2}$$

NOTE: The gradient of the v vs t graph could also be used.

## NEWTON'S FIRST LAW

| |
|---|
| A body at rest or moving at constant velocity will continue to do so unless acted upon by an external unbalanced force. |

**Example**


The force needed to move an object appears to be dependent on the mass.

The larger the mass, the greater the force required to move the object.

Newton's First Law is often referred to as the law of ***inertia***.

Inertia is considered to be the property of a body which, when still, makes it hard to move and, when moving, keeps it going.

This property seems to be determined by mass - the greater the mass, the greater the inertia.

**Example**
A train has a lot of inertia due to its large size. It takes a great deal of force to start it moving (if it's stationary) or stop it (if it's moving).

## FORCES IN EQUILIBRIUM

Newton's first law states that an object will remain stationary or will move at constant velocity unless there is a nett force acting on it.

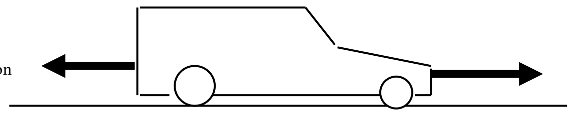
i.e. The forces are unbalanced.

Therefore, if the vector sum of all the forces acting on a body is zero, then the body is in a state of equilibrium where the forces balance.

i.e. $\boxed{\mathbf{\Sigma F = 0}}$

**Example**

A car moving along a straight and flat road maintains a constant velocity of 60.0 kmh$^{-1}$.




Since the car is maintaining a constant velocity, the nett or resultant force is zero.

i.e. $\Sigma F = 0$

=> <u>Force due to the engine = Force of friction acting</u>

# MOMENTUM

Momentum can be thought of as a measure of how hard it is to stop something that is moving.

e.g. A 1.00 tonne car moving at 10.0 $ms^{-1}$ is much easier to stop than a supertanker of $1.00 \times 10^5$ tonnes moving at the same speed.

The magnitude of momentum is given by:

$$\boxed{\mathbf{p = mv}}$$

where p = the momentum of the object ($kgms^{-1}$)
m = the mass of the object (kg)
v = the velocity of the object ($ms^{-1}$)

**Note:** Momentum is a vector and requires a direction to be given.

An object which changes velocity will change its momentum. This change in momentum is given by:

$$\Delta p = p_{final} - p_{initial}$$

$$= mv - mu$$

$$= m(v-u)$$

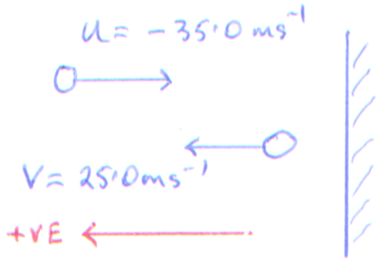
$$\therefore \qquad \boxed{\mathbf{\Delta p = m\Delta v}}$$

where $\Delta p$ = the change in momentum ($kgms^{-1}$)
$\Delta v$ = the change in velocity of the object ($ms^{-1}$)

**Example**
A young girl practises her tennis by hitting a ball against the practice wall at Carine open space. The 85.0 g ball strikes the wall at 35.0 $ms^{-1}$ and bounces off at 25.0 $ms^{-1}$. Calculate:

(a) the change in velocity of the ball.
(b) the change in momentum of the ball.

(Assume all velocities are horizontal to the ground just before and after impact.)




(a) $\Delta v = v - u$

$= 25{\cdot}0 - (-35{\cdot}0)$

$= 60{\cdot}0\ ms^{-1}$ away from the wall.

(b) $\Delta p = m\Delta v$

$= (0{\cdot}085)(60{\cdot}0)$

$= 5{\cdot}10\ kgms^{-1}$ away from the wall

## NEWTON'S SECOND LAW

| |
|---|
| The time rate of change of momentum of a body is directly proportional to the magnitude of the applied resultant force and is in the direction of that resultant force. |

$$\text{i.e.} \qquad \Sigma F \propto \frac{m\Delta v}{t}$$

$$\therefore \qquad \Sigma F = \frac{m\Delta v}{t}$$

where $\Sigma F$ = the sum of all forces acting on the body (N) (i.e. the resultant force)

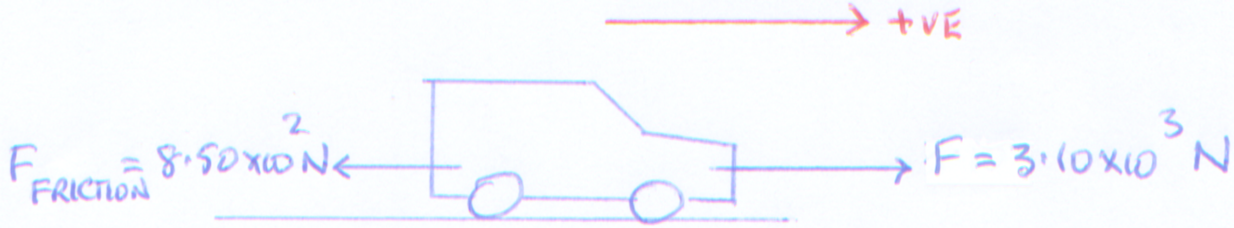
Given that $a = {}^{\Delta v}/_{t}$ :

$$\boxed{\mathbf{\Sigma F = ma}}$$

where $a$ = acceleration of the body in the direction of the resultant force ($ms^{-2}$)

**Example**

A car of mass $1.10 \times 10^3$ kg is accelerated by a motor which provides a force of $3.10 \times 10^3$ N. A frictional force of $8.50 \times 10^2$ N opposes the motion of the car. Calculate the acceleration of the car across the ground.




$$\Sigma F = ma$$

$$\Rightarrow 3{\cdot}10\times10^3 - 8{\cdot}50\times10^2 = (1{\cdot}10\times10^3)(a)$$

$$\Rightarrow \underline{a = 2{\cdot}04\ ms^{-2}\ \text{forwards}}$$

The equation for Newton's second law can be written as:

$$Ft = m\Delta v$$

By definition:

| |
|---|
| The impulse affecting an object is given by the product of the force acting and the time taken for the force to act. |

i.e. $\boxed{\mathbf{I = Ft}}$

where
- $I$ = the impulse experienced by the object (Ns)
- $F$ = the impulsive force acting (N)
- $t$ = the time taken for the force to act (s)

Combining the equations covered thus far gives the following.

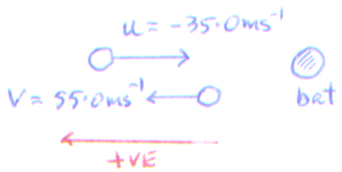
$$\boxed{\mathbf{I = Ft = m\Delta v = \Delta p}}$$

***** This is the best equation to remember and use! *****

**Example 1**

During a game of baseball, a batter strikes a ball thrown at him at 35.0 $ms^{-1}$. It leaves the bat at 55.0 $ms^{-1}$ after a contact time of 0.116 s. Given that the ball has a mass of 145 g, calculate:

(a) the change in velocity of the ball.
(b) the impulsive force acting.
(c) the impulse experienced by the ball.




(a) $\Delta v = v - u$

$= 55.0 - (-35.0)$

$= 90.0\ ms^{-1}$ away from the batter.

(b) $I = Ft = m\Delta v = \Delta p$ ← Choose the appropriate equations to use.

$\Rightarrow F = \dfrac{m\Delta v}{t}$

$= \dfrac{(0.145)(90.0)}{0.116}$

$= 1.12 \times 10^{2}$ N away from the batter.

(c) $I = Ft$

$= (1.12 \times 10^{2})(0.116)$

$= 13.0$ Ns away from the batter.

**Example 2 - AIR BAGS**
Air bags are designed to protect the occupants of cars during collisions.

As seen previously: $F = {}^{m\Delta v}/_{t} = {}^{\Delta p}/_{t}$

The head and body of a person will undergo a change in momentum ($\Delta p$). The value of $\Delta p$ can be considered constant - it is determined by the speed of the car at impact.

The air bag ***increases the time*** for $\Delta p$ to occur. Thus the force (F) experienced by the person is smaller.

***The greater the time taken for the collision to occur, the smaller the forces experienced by the occupants.***

**OTHER APPLICATIONS**

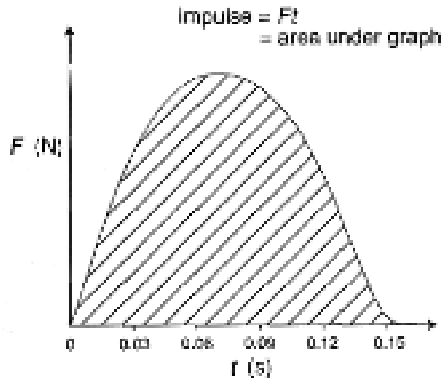
- **Crumple zones in cars** - these collapse in a controlled fashion. The change in momentum of the car occurs over a longer time so the forces experienced are smaller.

- **Sport shoes** - soft soles and "gel" increase the contact time of the foot with the ground.

- **Stunt bags / high jump bags / etc** - all designed to collapse or deform in a uniform manner. Again the time for $\Delta p$ to occur is increased.

- **Sand / rubber around children's playgrounds** - contact time with the ground is increased when a child falls from the equipment.

During a collision between two objects, the forces applied are ***not*** uniform. By graphing the change in F with time t, the area under the graph gives the value of the impulse (I).



## NEWTON'S THIRD LAW

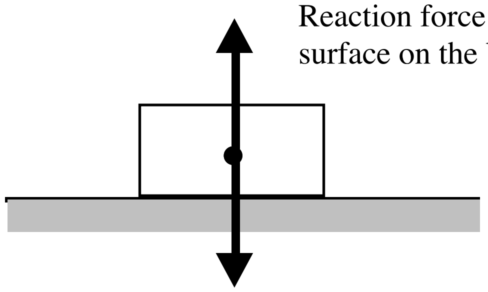
| |
|---|
| For every action there is an equal and opposite reaction. |

Newton realised that forces occur in pairs and act on different objects.

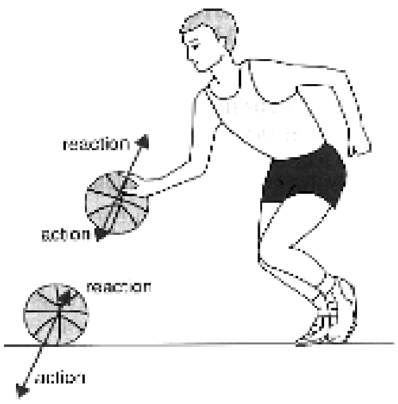
e.g. Consider a block sitting at rest on a table.

The force due to gravity (weight) acts downwards on the block. Thus the block exerts a force onto the table.

The table exerts an equal and opposite force to support it. The table exerts a force onto the block.

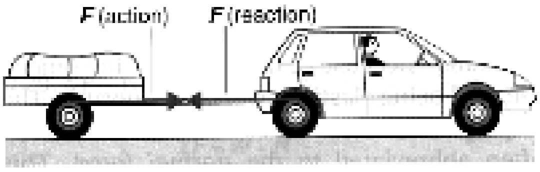



## Examples




The player pushes on the ball (action) and the ball pushes back equally on his hand (reaction).

When the ball hits the ground, the ball pushes on the ground (action) while the ground pushes back equally (reaction).

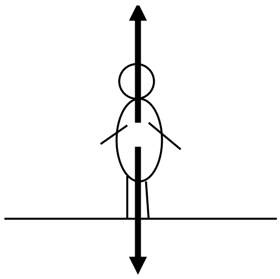



The car pulls on the trailer (action) and the trailer provides a reaction force on the car.

## APPARENT WEIGHT

When a person stands still on the ground, the weight they feel through their feet is equal to the reaction force exerted by the ground onto them.

This reaction force is called the ***apparent weight***. In this case the person's apparent weight is equal to their real weight.




When more than one force acts on a body, the resultant force is found as follows.

| | | | | |
|---|---|---|---|---|
| **RESULTANT FORCE** | **=** | **FORCES IN SAME DIRECTION AS THE NETT ACCELERATION** | **-** | **FORCES IN THE OPPOSITE DIRECTION** |

### Examples

1. In the situation given above, the nett acceleration of the person is zero.

i.e. $\Sigma F = 0$

Take down as positive.

$\Sigma F = F_w - R = 0$

$\Rightarrow \quad R = F_w$

$\therefore$ <u>The apparent weight = the real weight.</u>

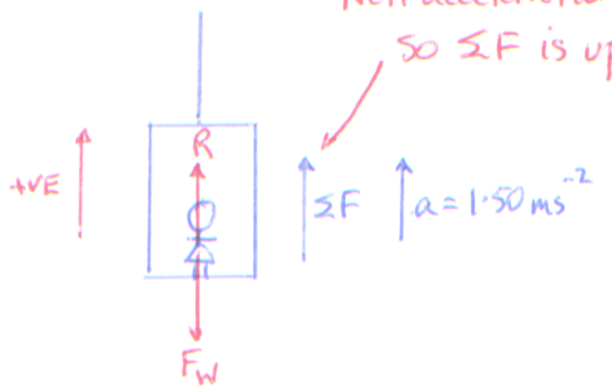
Decelerating as it moves down.

$\Rightarrow$ Acceleration is up.

## 2. Lift Problems

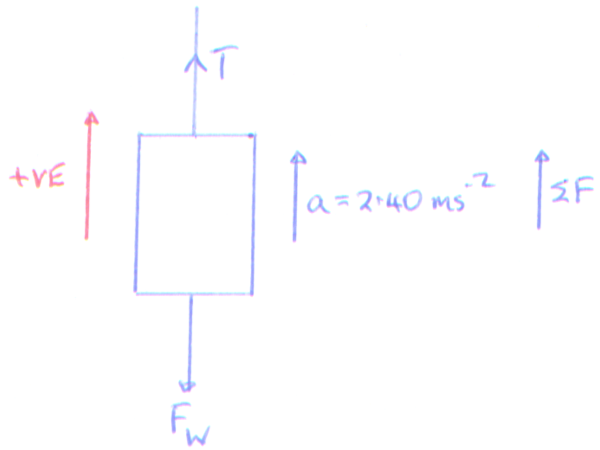
(a) A girl of mass 50.0 kg is in a lift which is decelerating at 1.50 $ms^{-2}$ as it nears the floor she has selected. What is the girl's apparent weight during this time?

Nett acceleration is up
so $\Sigma F$ is up.




$$\Sigma F = R - F_W$$
$$\Rightarrow R = \Sigma F + F_W$$
$$= ma + mg$$
$$= m(a+g)$$
$$= (50.0)(1.50 + 9.80)$$
$$= \underline{5.65 \times 10^2\ N}$$

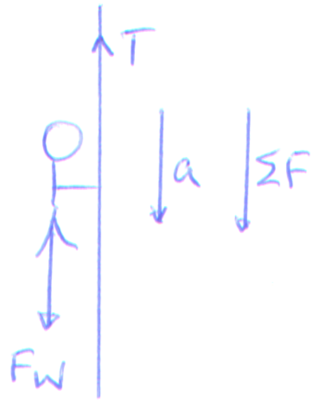
(b) The combined mass of the lift is 2.30 x $10^3$ kg. What is the tension in the cable if the lift is accelerating at 2.40 $ms^{-2}$ upwards as it rises from the basement?




$$\Sigma F = T - F_W$$
$$\Rightarrow T = \Sigma F + F_W$$
$$= ma + mg$$
$$= m(a+g)$$
$$= (2.30\times10^3)(2.40 + 9.80)$$
$$= \underline{2.81 \times 10^4\ N}$$

(c) A man of mass 80.0 kg is stuck on a ledge in a cave some 6.00 m above the cave floor. He has a rope which can only support a mass of 30.0 kg before it breaks.

He can use the rope to slide down to the floor if the rope is attached to the roof of the cave. Justiy this by calculation.




Maximum tension in the rope can just hold 30.0 kg

$\therefore T_{max} = (30.0)(9.80)$
$= 2.94 \times 10^2$ N

Man slides down so he will have a nett acceleration down.

To calculate nett acceleration:

$$\Sigma F = F_W - T$$
$$\Rightarrow ma = mg - T$$
$$\Rightarrow (80.0)(a) = (80.0)(9.80) - 2.94 \times 10^2$$
$$\Rightarrow a = 6.12 \text{ ms}^{-2}$$

$\therefore$ Man must slide down with a minimum acceleration of $6.12 \text{ ms}^{-2}$.

## CONSERVATION OF MOMENTUM

In any collision or interaction between two or more objects in an isolated system, the total momentum of the system will remain constant.

i.e. The total initial momentum = the total final momentum.

This can be expressed as follows.

$$\boxed{\Sigma \mathbf{p_i} = \Sigma \mathbf{p_f}}$$

where $\Sigma p_i$ = total initial momentum of all parts of the system
$\Sigma P_f$ = total final momentum of all parts of the system

**Note:** An isolated system is one on which no external unbalanced force is acting. This is very difficult to achieve on Earth because of gravity, friction and air resistance.

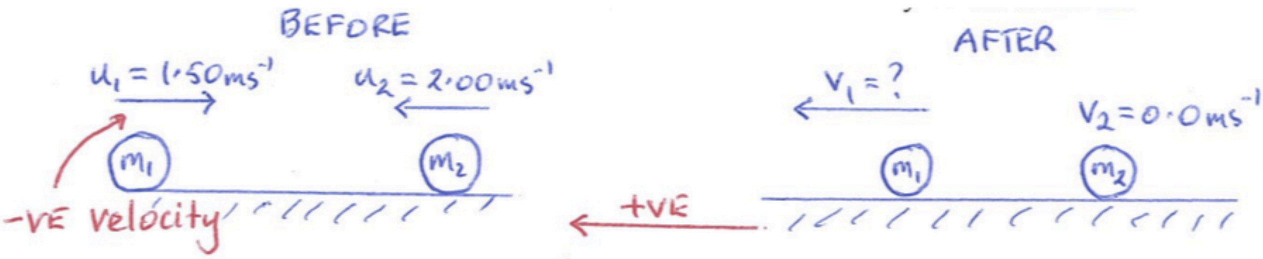
For the purposes of calculations, ***it will be assumed that all systems are isolated*** and that the only forces acting are the action/reaction forces during collision.

Also for the ease of calculation, ***only problems involving one dimension will be considered***, though it is relatively easy to consider vector additions in two dimensions.

**Examples**

1. **Collisions**

A billiard ball of mass 0.200 kg moving at 1.50 $ms^{-1}$ collides head-on with a similar ball moving at 2.00 $ms^{-1}$ in the opposite direction. If the second ball becomes stationary after the collision, determine the final velocity of the first ball.




$$\Sigma p_i = \Sigma p_f$$

$$\Rightarrow m_1 u_1 + m_2 u_2 = m_1 V_1 + m_2 V_2$$

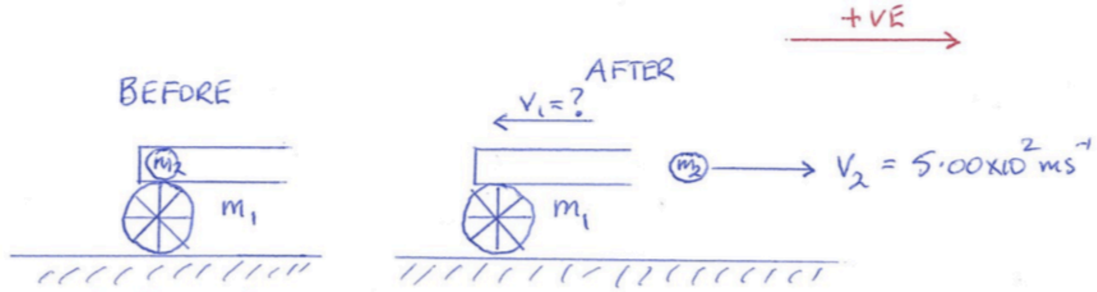
$$\Rightarrow (0{\cdot}200)(-1{\cdot}50) + (0{\cdot}200)(2{\cdot}00) = (0{\cdot}200)V_1 + 0$$

$$\Rightarrow \underline{V_1 = 0{\cdot}500 ms^{-1} \text{ backwards}}.$$

2. **Explosions**

A 9.50 x $10^2$ kg gun fires a 2.00 kg shell with a muzzle velocity of 5.00 x $10^2$ $ms^{-1}$. The gun recoils against a resisting force of 5.00 x $10^3$ N. Calculate:

(a) the recoil velocity of the gun.
(b) the recoil time of the gun.




Nothing is moving initially.

(a)

$$\Sigma p_i = \Sigma p_f$$

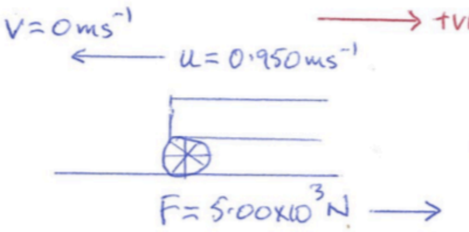
$$\Rightarrow m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$\Rightarrow 0 + 0 = (9.50 \times 10^2) v_1 + (2.00)(5.00 \times 10^2)$$

$$\Rightarrow v_1 = -0.950\ ms^{-1}.$$

$\therefore$ Recoil velocity = 0.950 $ms^{-1}$ backwards.

(b). The stopping of the gun is considered as a separate situation.




Choose this direction to be consistent to part (a).

Must calculate $\Delta v$ first.

$$\Delta v = v - u$$
$$= 0 - (-0.950)$$
$$= 0.950\ ms^{-1}$$ in the direction of the shell.

$$I = Ft = m\Delta v = \Delta p$$

$$\Rightarrow t = \frac{m\Delta v}{F} = \frac{(9.50 \times 10^2)(0.950)}{5.00 \times 10^3}$$

$$= 0.180\ s$$

## WORK

Work is defined as the product of the applied nett force and its displacement in the direction of the nett force.

i.e. $\boxed{\mathbf{W = Fs}}$

where W = the work done (J-joules)
F = the nett force applied to the object (N)
s = the displacement of the object (m)

Work is a ***scalar*** quantity - no direction is required.

When work is done on an object, it changes the energy of the object.

One ***JOULE*** of work is done when the point of application of a net force of one newton moves an object through a distance of one metre in the direction of the force.

If the force is applied at an angle to the direction of movement, it is *the **component of the force*** responsible for the work done.

**(This is Year 12 course work and will be covered then.)**

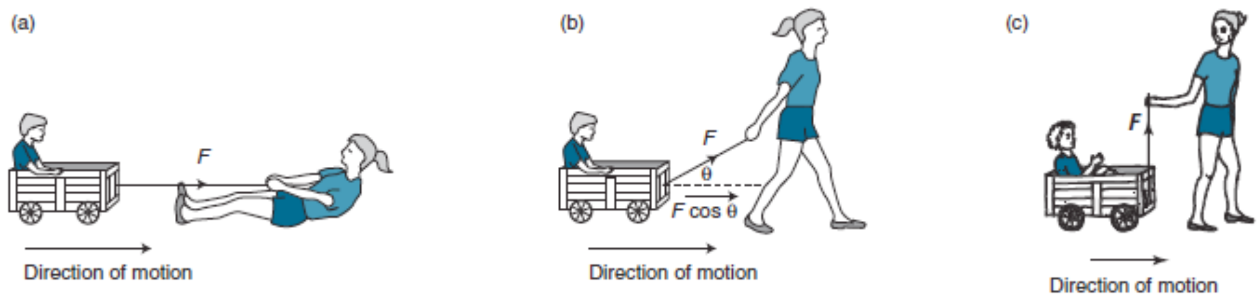



**Figure 4.9**

(a) If a force is applied in the direction of motion of the cart, then the force is at its most effective in moving the cart. (b) When the force is applied at an angle θ to the direction of motion of the cart, the force is less effective. The component of the force in the direction of the displacement, *F*cosθ, is used to calculate the work. (c) When the angle at which the force is acting is increased to a right angle (θ = 90°), then the component of the force in the direction of the intended displacement is zero and it does no work on the cart—provided of course that it doesn't lift the cart, in which case work would also be done against gravity.

If a force is applied to an object and it does not move, then no work has been done.

e.g. 1. A person may push continuously against a parked car until exhausted, but if the car does not move, ***no mechanical work has been done***.

However, the person has expended a significant amount of work internally (physiologically)!

**2.** Carrying a bag at a constant speed across the ground at a constant height has no work done on it.

Reason: the energy of the bag has not changed.
(See the section on Energy.)

## Examples

1. Calculate the work done against gravity by an athlete of mass 60 kg competing in the Rialto Tower Run-up illustrated in Figure 4.10. Use $g = 9.8\ \mathrm{m\ s^{-2}}$.

### Solution

Only the weight force need be considered in this example as the work in the vertical direction is all that is required.

$m = 60$ kg, $g = 9.8\ \mathrm{m\ s^{-2}}$, $s = \Delta h = 242$ m

Force required = weight = $mg = 60 \times 9.8 = 588$ N

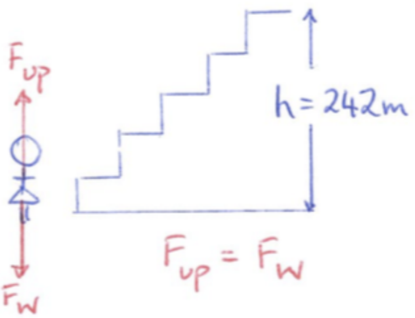
$W = Fs$

$= 588 \times 242$

$= 142\,296\ \mathrm{N\ m} = 1.4 \times 10^5\ \mathrm{J}$

**Figure 4.10**
The annual Rialto Tower Run-up is a race to climb 1254 steps up the Rialto Tower building in Melbourne. Contestants climb a vertical distance of approximately 240 m and the winner takes around 7 minutes to complete the race.

Consider movement vertically.




$$
\begin{aligned}
W &= F_{up} s \\
&= F_W s \\
&= mgs \\
&= (60.0)(9.80)(242) \\
&= \underline{1.42 \times 10^5\ \mathrm{J}}.
\end{aligned}
$$

2. The girl in Figure 4.9 pulls the cart by applying a force of 50 N at an angle of 30° to the horizontal. Assuming a force due to friction of 10 N is also acting on the wheels of the cart, calculate the work done if the cart is moved 10 m along the ground in a straight line.

### Solution

$F_1 = 50$ N, $\theta = 30°$, $F_f = 10$ N, $s = 10$ m

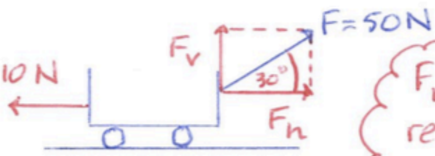
$\Sigma F = F_1 \times \cos 30° - F_f$

$= 50 \times 0.866 - 10 = 33.3$ N

Now $W = \Sigma Fs$,

and work $= 33.3 \times 10 = 330$ J

Done in Year 12 course.




$F_h$ is the force responsible for horizontal movement.

$$
\begin{aligned}
F_h &= 50 \cos 30° \\
&= 43.3\ \mathrm{N}.
\end{aligned}
$$

Horizontally: $\longrightarrow$ +VE

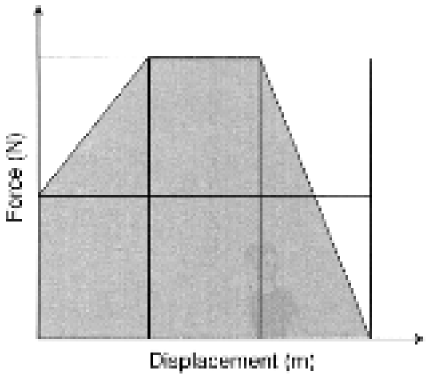
$$
\begin{aligned}
\Sigma F_h &= 43.3 - 10.0 \\
&= 33.3\ \mathrm{N}.
\end{aligned}
$$

$$
\begin{aligned}
W &= \Sigma F_h s \\
&= (33.3)(10.0) \\
&= \underline{333\ \mathrm{J}}.
\end{aligned}
$$

**Force - Displacement Graphs**

The area under such a graph gives a measure of the work done by the force.

This is particularly useful in situations where a force changes with displacement, such as in collisions.




The area under the graph $= W_{done}$

$= F_{ave}\ s$

Thus it is possible to determine the average force which acted during the situation being studied.

# ENERGY

Energy is a measure of a body or system's ability to do work.

There are various types of mechanical energy which are applied to movement.

**Kinetic Energy:** This is the energy possessed by a body when it is moving.

It is given by:

$$\boxed{E_k = \tfrac{1}{2}mv^2}$$

where $E_k$ = the kinetic energy (J)
$m$ = the mass of the object (kg)
$v$ = the velocity of the mass ($ms^{-1}$)

If a force acts on a nett body, it does work on it. In doing so there is a change in the kinetic energy of the body.

i.e. $$\boxed{W = \Delta E_k = \tfrac{1}{2}mv^2 - \tfrac{1}{2}mu^2}$$

where $^1/_2 mv^2$ = the final kinetic energy of the body.
$^1/_2 mu^2$ = the initial kinetic energy of the body.

**Note:** If the final $E_k$ is smaller than the initial $E_k$ of the object because it is slowing down, reverse the equation to produce a positive vale for $\Delta E_k$.

i.e. $W = \Delta E_k = {}^1/_2\, mu^2 - {}^1/_2\, mv^2$

Like all forms of energy, $E_k$ is a scalar and has no direction associated with it.

**Example**
A car of mass 1.30 x $10^3$ kg is moving along a road at 70.0 $kmh^{-1}$ when it brakes sharply to avoid a person on a skateboard coming out of a driveway. The speed is reduced to 30.0 $kmh^{-1}$ as the car swerves around the person. Calculate the work done by the brakes on the car.

$u = 70.0\ kmh^{-1}$
$= 19.44\ ms^{-1}$

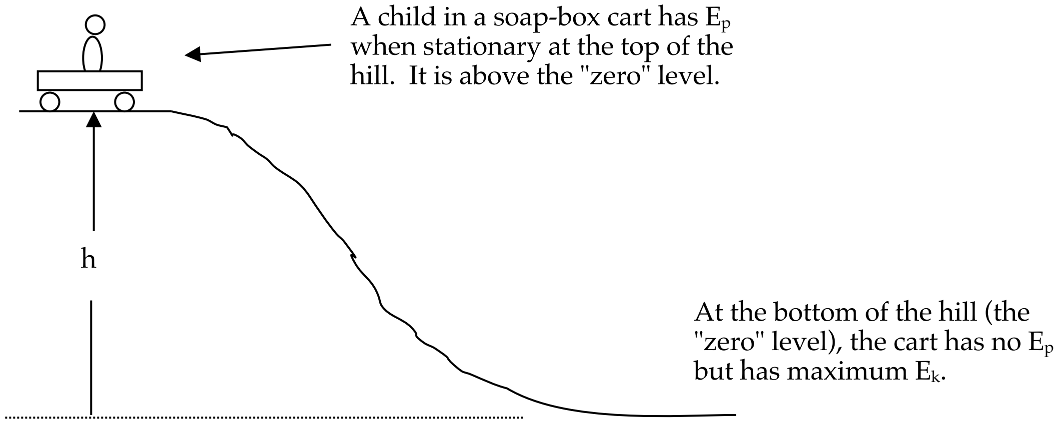
$v = 30.0\ kmh^{-1}$
$= 8.333\ ms^{-1}$

$$W = \Delta E_k = \tfrac{1}{2}mu^2 - \tfrac{1}{2}mv^2$$
$$= \tfrac{1}{2}m(u^2 - v^2)$$
$$= \tfrac{1}{2}(1.30\times10^3)\left[(19.44)^2 - (8.333)^2\right]$$
$$= 2.005\times10^5\ J$$

$\therefore \underline{W_{done} = 2.00\times10^5\ J}$

**Gravitational Potential Energy:** This is the energy possessed by a body due to its height above a given level.

Any object above a given "zero" level has the ability to release this energy as $E_k$ as it falls to this "zero" level.




The amount of potential energy is given by:

$$\boxed{E_p = mgh}$$

where
$E_p$ = the potential energy of the body (J)
m = the mass of the body (kg)
g = the acceleration due to gravity ($ms^{-2}$)
h = the height above the "zero" level (m)

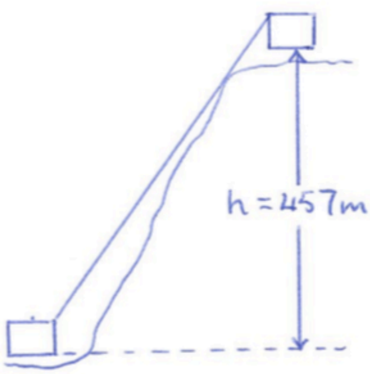
If an object is raised up to a height in the Earth's gravitational field, work must be done against the force of gravity.

This work is given by:

$$\boxed{W = \Delta E_p = mg\Delta h}$$

where $\Delta h$ = the height through which the object is raised.

**Example**
A cable car of combined mass 4.30 x $10^3$ kg is used to transport 22 people from the carpark to the top of Grouse Mountain in Vancouver, a vertical rise of 457 m. Calculate the work done by the motors operating the car.

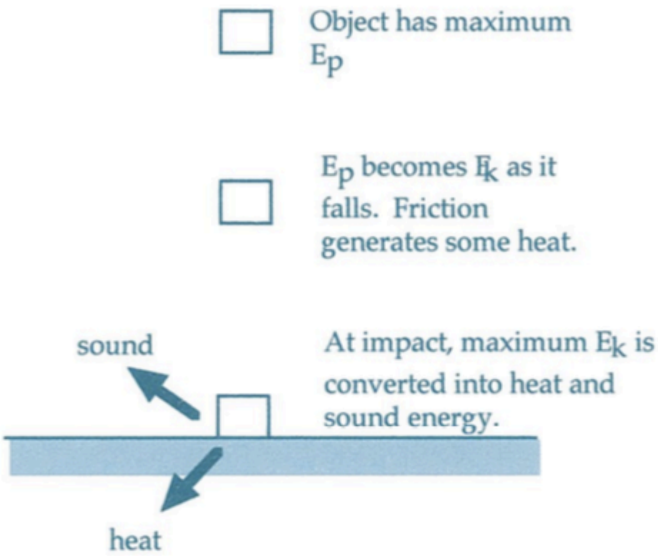



$$W_{done} = \Delta E_p = mg\Delta h$$
$$= (4{\cdot}30\times10^3)(9{\cdot}80)(457)$$
$$= 1{\cdot}926\times10^7 \text{ J}$$
$$\therefore \underline{W_{done} = 1{\cdot}93\times10^7 \text{ J}}$$

## CONSERVATION OF MECHANICAL ENERGY

In all situations, energy must be conserved - it can be transformed from one form to another, but the total amount stays the same.

**Example**
Consider an object dropped from a height onto a hard surface.

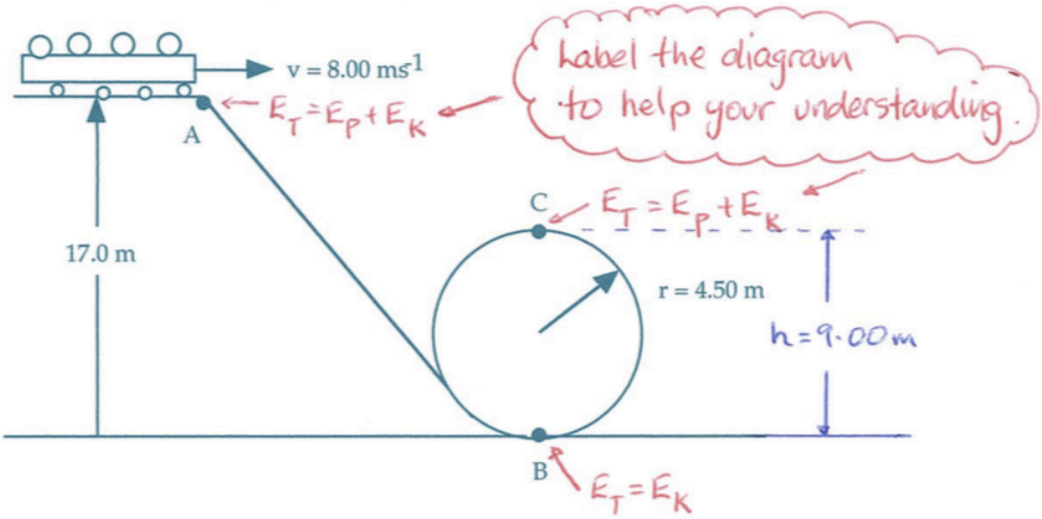



The above situation describes a particular case where the total mechanical energy is conserved.

$$\begin{aligned} E_{total} &= E_p + E_k \\ &= mgh + \tfrac{1}{2}mv^2 \end{aligned}$$

**Note:** ***In calculations, it will generally be considered that friction is not acting.***

**Example**
A roller coaster of mass $3.70 \times 10^3$ kg is moving at 8.00 $ms^{-1}$ at point A as shown below.




(a) Determine the total energy of the coaster at A.

(b) Calculate the speed achieved at point B.

(c) Does the coaster safely make it through the loop if the minimum speed to do so is 6.64 $ms^{-1}$? Justify your answer by calculation.

NOTES
Stationary at a height $\Rightarrow E_T = E_P$
Moving at a height $\Rightarrow E_T = E_P + E_K$
Moving at lowest level $\Rightarrow E_T = E_K$

(a) At A: Moving at a height

$\therefore E_T = E_P + E_K$
$= mgh + \frac{1}{2}mv^2$
$= (3.70\times10^3)(9.80)(17.0) + \frac{1}{2}(3.70\times10^3)(8.00)^2$
$= 7.348 \times 10^5$ J

$\therefore \underline{E_T = 7.35 \times 10^5 \text{ J}}$

(b) At B: Moving at lowest level

$\therefore E_T = E_K = \frac{1}{2}mv^2$
$\Rightarrow 7.348\times10^5 = \frac{1}{2}(3.70\times10^3)v^2$
$\Rightarrow v = 19.93\ ms^{-1}$

$\therefore \underline{\text{Speed} = 19.9\ ms^{-1}}$

(c) At C: Moving at a height.

$\therefore E_T = E_P + E_K$
$= mgh + \frac{1}{2}mv^2$
$\Rightarrow 7.348\times10^5 = (3.70\times10^3)(9.80)(9.00) + \frac{1}{2}(3.70\times10^3)v^2$
$\Rightarrow v = 14.86\ ms^{-1}\ (> 6.64\ ms^{-1})$

$\therefore$ The coaster safely negotiates the loop.

## ELASTIC AND INELASTIC COLLISIONS

In all collisions, momentum is conserved. So too is energy (in all its different forms). However, it is unusual for a particular type of energy to be conserved during a collision.

**Elastic Collision** - one where no kinetic energy is lost, and the $E_k$ and momentum of the system are both conserved.

e.g. In the interaction between atoms and sub-atomic particles.

Elastic collisions do not exist in everyday situations - only the sub-atomic.

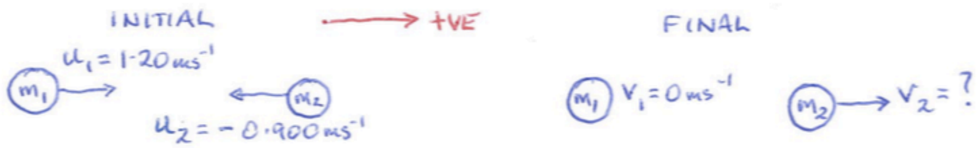
**Inelastic Collision** - one where momentum is conserved but not $E_k$ (which is transformed into other forms of energy).

e.g. Collisions between billiard balls, cars, football players, etc.

**Example**
A billiard ball of mass 0.150 kg moving at 1.20 ms$^{-1}$ collides head-on with a similar ball moving at 0.900 ms$^{-1}$ in the opposite direction. After the collision, the first ball becomes stationary.

(a) Calculate the final velocity of the second ball.

(b) Compare the sum total of the initial and final kinetic energies.




(a) Momentum is conserved in mechanical systems.

$$\Sigma p_i = \Sigma p_f$$

$$\Rightarrow m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$\Rightarrow (0{\cdot}150)(1{\cdot}20) + (0{\cdot}150)(-0{\cdot}900) = 0 + (0{\cdot}150)v_2$$

$$\Rightarrow v_2 = 0{\cdot}300\,\text{ms}^{-1}$$

$\therefore$ $\underline{v_2 = 0{\cdot}300\,\text{ms}^{-1} \text{ in the direction of ball 1.}}$

(b) $\Sigma E_k(\text{initial}) = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2$

$= \frac{1}{2}(0{\cdot}150)(1{\cdot}20)^2 + \frac{1}{2}(0{\cdot}150)(0{\cdot}900)^2$

$= 0{\cdot}1688\,\text{J}$

$\Sigma E_k(\text{final}) = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$

$= \frac{1}{2}(0{\cdot}150)(0) + \frac{1}{2}(0{\cdot}150)(0{\cdot}300)^2$

$= 6{\cdot}75 \times 10^{-3}\,\text{J}$

$\therefore$ $\underline{\Sigma E_k(\text{initial}) \neq \Sigma E_k(\text{final})}$

## POWER

**Power** - the rate at which energy is transformed, or the rate at which work is done.

$$\boxed{P = \frac{W}{t} = \frac{E}{t}}$$

where P = the power generated (W - Watts)
W = the work done (J)
E = the energy transformed (J)
t = the time taken for the process (s)

Calculating the power developed is straightforward when there is a change in mechanical energy or work, but some situations involve ***movement at a constant speed*** which involves no change in mechanical energy.

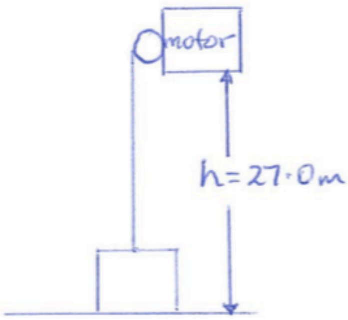
Using W = Fs, the equation changes as follows.

$$\boxed{\begin{aligned} P &= \frac{W}{t} = \frac{Fs}{t} \\ &= F\,v_{av} \end{aligned}}$$

where F = average force applied (N)
$v_{av}$ = the average velocity of the object ($ms^{-1}$)

**Examples**

1. An electric motor drives a lift of nett mass $1.47 \times 10^4$ kg through a height of 27.0 m in 12.0 s. What is the power of the motor?




$$P = \frac{W}{t} = \frac{\Delta E_p}{t} = \frac{mg\Delta h}{t}$$

$$= \frac{(1.47\times10^4)(9.80)(27.0)}{12.0}$$

$$= 3.241\times10^5 \text{ W}$$

$$\therefore \underline{P = 3.24\times10^5 \text{ W}}$$

2. A cyclist in a road race is producing an average of 5.00 x $10^2$ W of power whilst moving at 48.0 $kmh^{-1}$ along a flat road. Ignoring any air resistance, what average force is being supplied through the pedals?

$V_{const} = 48.0\ kmh^{-1}$

$= 13.33\ ms^{-1}$

$P = \frac{W}{t} = \frac{Fs}{t} = F v_{ave}$

$\Rightarrow F = \frac{P}{V_{ave}}$

$= \frac{5.00 \times 10^2}{13.33}$

$= 37.51\ N$

$\therefore \underline{F = 37.5\ N \text{ forwards}}$